

10-Mark Questions & Answers — DBMS Module 5: Query Optimization & Transaction Concurrency Control

Q1. Explain query optimization in detail (Heuristic evaluation trees, relational equivalences, cost estimation formulas, join selection).

Detailed breakdown of:

1. Parsing SQL into Initial Relational Algebra Expression Tree.
2. Applying Equivalence Rules (Pushing σ and π down).
3. Cost estimation for Block Nested Loop, Hash Join, and Merge Join.
4. Physical Plan Selection based on Catalog Statistics (n_r , b_r , $V(A,r)$).